Shruti kamnani -25070521147

SECTION-C1

EXPERIMENT-8.1.1 MULTIPLICATION OF TWO SQUARE MATRICES

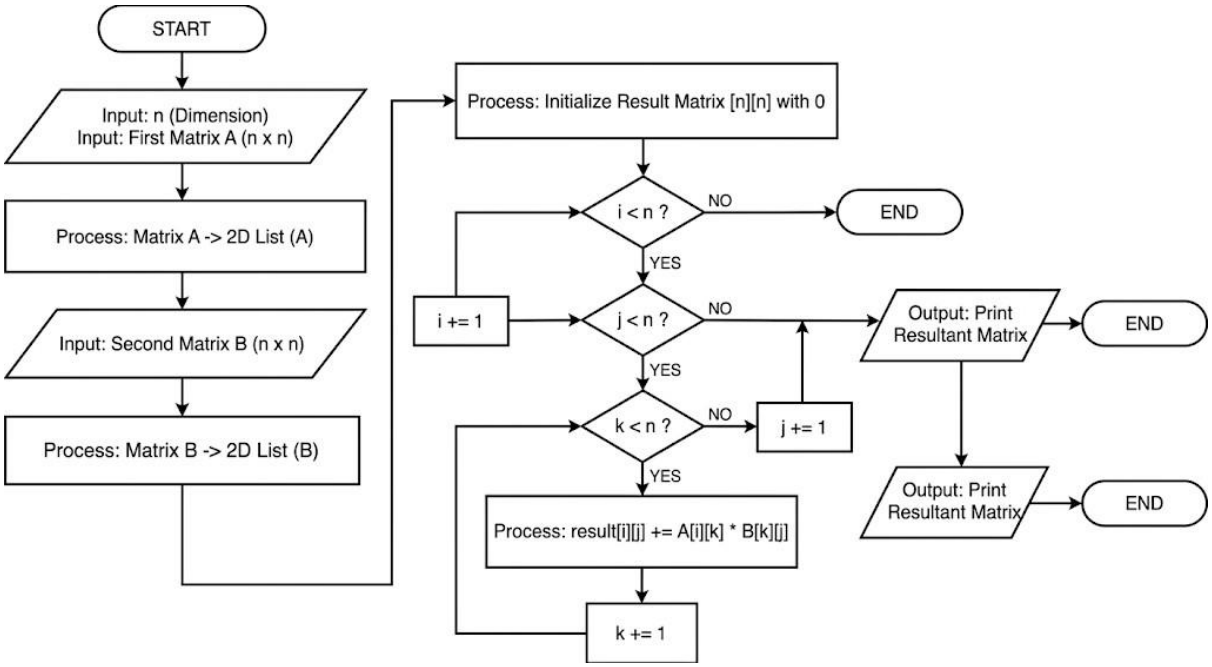
ALGORITHM:

1. Take size n.
2. Input two matrices
3. For each position:

- Multiply row of first matrix with column of second matrix
- Add the results

4. Store the value
5. Print the final matrix

FLOWCHART:



CODE:

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shruti.kamnani.batch2025@sitnagpur.siu.edu.in Support Logout

8.1.1. Multiplication of Two Square Matrices

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases +

Multiplica... Submit

```python
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    A.append(list(map(int, input().split())))
print("second matrix:")
B = []
for i in range(n):
    B.append(list(map(int, input().split())))
    result = [[0]*n for _ in range(n)]
for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j] += A[i][k] * B[k][j]
print("Resultant Matrix:")
for r in result:
```

Average time 0.018 s 18.25 ms

Maximum time 0.021 s 21.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 19 ms Debug

Expected output	Actual output
dimension: 2	dimension: 2
first matrix:	first matrix:
1 2	1 2
3 4	3 4
second matrix:	second matrix:
5 6	5 6
7 8	7 8
Resultant Matrix:	Resultant Matrix:
19 22	19 22
43 50	43 50

Terminal Test cases

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